

Cycle-Covering Dynamics of a Three-State Nonlinear Kernel

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Abstract

Working within the established framework of network quotients, graph fibrations, equitable partitions, and synchrony subspaces [SGP03, DL15, GR01], we study a specific discrete complex coupled-map model and make one embedding relationship exact. We study a three-state nonlinear oscillator map — a Mexican-hat amplitude nonlinearity coupled through the Laplacian of the triangle graph, followed by a harmonic-3 phase-synchronization step — and show that it arises as an **exact invariant reduction** of a larger periodic system on a cycle graph. The mechanism is a graph covering. We prove that for any divisor $d \mid M$ the residue map $p : C_M \rightarrow C_d$, $p(n) = n \bmod d$, induces a pull-back $P : \mathbb{C}^d \rightarrow \mathbb{C}^M$ that intertwines the cycle Laplacians ($\Delta_M P = P \Delta_d$), and that in fact $\Delta_M P = P \Delta_d \iff d \mid M$ for all positive M, d . On the pulled-back subspace $V_d = \text{im } P$ the operator Δ_M **equals** Δ_d in the scaled residue-indicator basis (via the isometry $Q = P/\sqrt{M/d}$, $Q^* \Delta_M Q = \Delta_d$). We give the complete invariance classification: within $1 \leq d \leq M$, V_d is Δ_M -invariant iff $d \mid M$ or $(M, d) = (3, 2)$. Specializing to $(M, d) = (12, 3)$ gives a four-sheeted graph covering with deck group $\mathbb{Z}_4$ and $\Delta_{12} \upharpoonright_{V_3} = L_3$, the triangle Laplacian; $V_3 = \text{span}\{e_0, e_4, e_8\}$. We prove that a natural M -ring nonlinear extension with three-periodic coefficients satisfies $F_M(P\Omega) = P F_3(\Omega)$ **for every M divisible by three**, under an explicit zero-amplitude phase convention; a numerical cross-check confirms the exact one-step identity to floating-point rounding, independently reproduced, while long trajectories in unrestricted regimes are shown *not* to remain machine-close despite the exact algebra. Linearizing transverse to V_3 (for $M = 12$) via the real deck-character decomposition $\mathbb{R}^{24} = U_0 \oplus U_2 \oplus U_{13}$, we exhibit linearly stable and unstable fixed and periodic regimes with their normal Floquet multipliers; direct nonlinear perturbation experiments confirm the predicted normal decay and growth rates in representative regimes. We record the exact-algebra coincidence with a second, independently defined ring operator, and a **sufficient** compatibility condition under which a k -only scalar feedback preserves the reduction. Strongest conclusion: **for every M divisible by three the specified three-state nonlinear map admits an exact invariant pull-back realization on the M -cycle; for the $M = 12$ realization studied numerically the invariant sector exhibits linearly stable and unstable transverse regimes, with direct perturbations confirming transverse attraction in representative linearly stable regimes**. The reduction machinery is standard network-quotient / synchrony-subspace theory; the contribution is the specific model together with its explicit deck-resolved transverse-stability atlas (§1.4). No novelty or priority is claimed.

Keywords: cycle graph, graph covering, discrete Laplacian, invariant subspace, pull-back, deck group, normal Floquet multiplier, nonlinear map reduction.

1. Introduction

A recurring situation in coupled-oscillator networks is that a small, symmetric subsystem behaves as though embedded in a larger, more regular one. This paper makes one such relationship exact. We begin from a concrete three-state map $F_3 : \mathbb{C}^3 \rightarrow \mathbb{C}^3$,

$$F_3(\Omega) = S_3(\Omega + \varepsilon \Omega \odot (k - |\Omega|^2) + g L_3 \Omega), \quad L_3 = \begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix},$$

where $\odot$ is entrywise multiplication, $|\Omega|^2$ is entrywise, $k \in \mathbb{R}^3$ is a fixed **amplitude coefficient** triple (no wavenumber or physical meaning is assigned to it), $\varepsilon, g \in \mathbb{R}$, and S_3 is the phase-only synchronization operator (defined precisely in §6.1 via the zero-amplitude convention below),

$$\phi_j \mapsto \phi_j + \lambda \sum_{i \neq j} \sin(3(\phi_i - \phi_j)), \quad \Omega_j = r_j e^{i\phi_j},$$

which leaves each amplitude r_j unchanged. The matrix L_3 is simultaneously the Laplacian of the triangle graph C_3 and of the complete graph K_3 (they coincide on three vertices); its spectrum is $\{0, -3, -3\}$ in the sign convention $(\Delta_M f)(n) = f(n+1) + f(n-1) - 2f(n)$ used throughout.

The central question is structural: **is F_3 the shadow of a system on a larger, more uniform graph?** We answer yes, with C_M (any M divisible by three) as the larger graph, and we make “shadow” precise as a graph covering together with the pull-back of functions along it. The $M = 12$ realization is then studied numerically for transverse stability.

Contributions:

1. The **pull-back intertwining theorem** for cycle Laplacians (§3): $\Delta_M P = P \Delta_d \iff d \mid M$ (all positive M, d), and $Q^* \Delta_M Q = \Delta_d$; with the complete invariance classification of $V_d = \text{im } P$.
2. The precise **graph-theoretic status** of the $12 \rightarrow 3$ relationship (§4): a four-sheeted covering, deck group $\mathbb{Z}_4$, quotient $C_{12}/\langle n \mapsto n+3 \rangle \cong C_3$ — not a subgraph inclusion.
3. The **Fourier and symmetry decomposition** (§5): $V_3 = \text{span}\{e_0, e_4, e_8\}$, exact complement spectrum, and the image of the C_{12} dihedral action as the S_3 symmetry of L_3 .
4. A **nonlinear extension** (§6) proved to satisfy $F_M(P\Omega) = P F_3(\Omega)$ for every $M = 3q$, with an explicit zero-amplitude convention, verified numerically and independently reproduced.
5. A **transverse-stability analysis** (§7, $M = 12$) using the real deck decomposition, distinguishing linear spectral stability from nonlinear attraction, with representative regimes and (§8) a **direct perturbation experiment** confirming the linear prediction.
6. Two structural addenda: the **exact-algebra coincidence** with a second ring operator (§9) and a **sufficient feedback-compatibility** condition (§10).

Sections 3–5 are elementary and self-contained. Sections 6–8 mix proved invariance with carefully delimited numerical fact. Scope is deliberately conservative: the ring extension is a construction of this paper, and transverse attraction is local and regional, not global (§12).

1.2 Established frameworks (background)

The constructions below are standard in network dynamics, and we cite them as background rather than as new results. The invariant sector V_3 is a **synchrony subspace** (polydiagonal) of a **balanced / equitable partition** of the cycle [SGP03, GST05, GS06]; the reduction $C_M \rightarrow C_3$ is a **graph covering**, equivalently a surjective **graph fibration** [BV02, DL15, NRS16]; the Laplacian identity of Theorem 1 is the **equitable-partition / quotient-(divisor-)matrix identity** of algebraic graph theory [GR01, OYSB13]; and the transverse-stability analysis of §7 uses the **master-stability** framework [PC98] with the symmetry-adapted (irreducible-representation)

cluster decomposition of [Pecora14, Sorrentino16]. For the specific regular cycle covers considered here these descriptions coincide: covering $\equiv$ surjective fibration $\equiv$ balanced-coloring quotient $\equiv$ equitable-partition quotient (a synthesis of the groupoid / fibration / balanced-coloring pictures is [GFBC24]). Accordingly the **divisor-case quotient construction** in Theorem 1 (§3) is a cycle specialization of the standard equitable-partition and network-quotient framework [GR01, OYSB13, SGP03, DL15]; the exact converse ($\Delta_M P = P \Delta_d \Rightarrow d \mid M$) and the invariance classification (including the exceptional (3,2) case) are **calculated here**, and no novelty or priority is claimed. The Fourier / deck spectral decomposition of §5 is standard lift / quotient spectral theory [GR01, DF19].

1.3 Relation to prior work

- **Network quotients and synchrony subspaces.** The subspace $V_3 = \{x : x_n = x_{n+3}\}$ is the polydiagonal of the balanced equivalence relation grouping the M ring sites into three residue classes. Balanced synchrony and quotient dynamics are established for admissible vector fields [SGP03, GST05, GS06], with an explicit smooth-network-map formulation in [NRS16]. The V_3 used here is one such synchrony subspace; Proposition 2 verifies compatibility for the particular composed map considered here.
- **Graph fibrations.** The residue map $p : C_M \rightarrow C_3$ is a surjective graph fibration [BV02], and a fibration relates the network dynamics on total and base graph [DL15, NRS16]. Here the pull-back $P : \mathbb{C}^3 \rightarrow V_3 \subset \mathbb{C}^M$ is a linear **bijection from the base space onto V_3** , so the intertwining $F_M \circ P = P \circ F_3$ (Proposition 2) **identifies the base dynamics F_3 with the restriction $F_M \upharpoonright_{V_3}$** . We state it this way, rather than as a loose “conjugacy,” because P is a bijection onto the invariant sector but not onto all of $\mathbb{C}^M$; the identification is of F_3 with F_M restricted to V_3 .
- **Equitable partitions and quotient Laplacians.** The residue partition is equitable; $Q^* \Delta_M Q = \Delta_3$ (Theorem 1(iii)) is the quotient-(divisor-)matrix identity, and the eigenvalue inheritance (Theorem 1(iv)) is the standard fact that quotient-matrix eigenvalues form a sub-multiset of the graph's [GR01, OYSB13, Schaub16].
- **Transverse stability.** The tangential / transverse split $U_0 \oplus (U_2 \oplus U_{13})$ and the block-diagonalization of the normal variational dynamics by deck characters (§7) are the abelian ($\mathbb{Z}_4$) case of the symmetry-adapted / IRR cluster decomposition [Pecora14, Sorrentino16], built on the master-stability framework [PC98]. Cluster variational stability is treated in [Sorrentino16], and general normal stability including periodic orbits in [ABS96]; the periodic-orbit Floquet calculations below are carried out for this map.

1.4 Contribution (what is model-specific)

Given the background above, the contribution is the specific worked instance, not the reduction machinery. The following are model-specific:

1. **The map itself** — a discrete complex state $\Omega \in \mathbb{C}^M$ combining a Mexican-hat amplitude nonlinearity $\varepsilon \Omega \odot (k - |\Omega|^2)$, cycle-Laplacian coupling $g \Delta_M \Omega$, and a *separate* harmonic-3 phase-synchronization step with three-periodic k . We did not locate a near-identical model in a focused literature search (§6.5).
2. **The explicit all-3 $\mid M$ reduction** for this coupling, written out with the zero-amplitude convention (§6), and the one-step numerical cross-check.
3. **The deck-resolved transverse-stability atlas** of the $M = 12$ realization (§7–§8): the specific stable / unstable fixed and periodic regimes, the crossing $g_* = 0.4220744431784353$, the specific normal Floquet values, and the direct-perturbation confirmation.
4. **The linear-vs-nonlinear stability delimitation** for this map (§7.2), with the explicit algebraic-decay and transient-growth counterexamples.
5. **The shared- L_3 -sector observation** relating the map to a second Δ_{12} -based (“RSB”) operator (§9), as a statement of shared quotient / equitable-partition algebra only.

6. **The k -only feedback compatibility criterion** (§10), a sufficient pull-back condition specialized to this architecture.

These are what a specific worked example contributes. We make **no claim of novelty, priority, or a new theorem or mechanism**: every construction above has a standard home in the cited literature, and the model-specific items are computations and constructions *of this model* rather than new general theory (see §12 and PAPER_A_LITERATURE_CONTEXT_v0.1.md).

2. Cycle graphs and pull-back spaces

Fix $M \geq 1$. Let C_M be the cycle graph on $\mathbb{Z}_M = \{0, \dots, M-1\}$, edges $\{n, n+1\} \pmod{M}$; C_2 is the doubled edge and C_1 a single loop (a dart/incidence description for these degenerate sizes is given in §4.3). The **cycle Laplacian** $(\Delta_M f)(n) = f(n+1) + f(n-1) - 2f(n)$ is real symmetric circulant; note $\Delta_1 = (0)$ and $\Delta_2 = \begin{pmatrix} -2 & 2 \\ 2 & -2 \end{pmatrix}$ are the degenerate cases, and the generic first row $(-2, 1, 0, \dots, 0, 1)$ describes Δ_M for $M \geq 3$. Its eigenvectors are the Fourier modes $e_j(n) = e^{2\pi i j n / M}$, $j \in \mathbb{Z}_M$, with

$$\Delta_M e_j = -4 \sin^2(\pi j / M) e_j. \quad (2.1)$$

In particular $\Delta_3 = L_3$ with spectrum $\{0, -3, -3\}$.

Pull-back for all positive M, d . Define $P : \mathbb{C}^d \rightarrow \mathbb{C}^M$ by

$$(Pu)_n = u_{n \bmod d}, \quad V_d := \text{im } P,$$

i.e. the $M \times d$ $\{0, 1\}$ -matrix with $P_{n, n \bmod d} = 1$. **We do not identify V_d with $\{f : f(n+d) = f(n)\}$ outside the divisor case**: for $1 \leq d \leq M$, those two spaces coincide iff $d \mid M$; no shift-fixed characterization is asserted here for $d > M$. (Within $1 \leq d \leq M$: the shift-fixed space has dimension $\gcd(M, d)$ and is always Δ_M -invariant, whereas $\text{im } P$ has dimension $\min(M, d)$ and need not be; e.g. at $(M, d) = (5, 3)$, $\text{im } P$ has dimension 3 and is not invariant while $\{f(n+3) = f(n)\}$ has dimension 1 and is invariant. The unrestricted “only when $d \mid M$ ” wording is avoided because it fails for $d > M$ — e.g. $(M, d) = (2, 4)$, where $\text{im } P = \mathbb{C}^2$ and the wrap-by-4 shift is the identity on $\mathbb{C}^2$, so both spaces are $\mathbb{C}^2$ although $4 \nmid 2$.) For $d \mid M$, $q = M/d$, distinct columns of P are orthogonal indicators of the d residue classes, each of squared norm q , so

$$P^*P = q I_d, \quad Q := P/\sqrt{q} \text{ satisfies } Q^*Q = I_d, \quad (2.2)$$

Q an isometry onto V_d and QQ^* the orthogonal projector onto V_d . (Here Q^* denotes the Hermitian adjoint; Q is real, so $Q^* = Q^\top$ — we write Q^* to signal the complex inner product.) The pull-back scales norm by

$$\|P\Omega\| = \sqrt{M/d} \|\Omega\|, \quad \text{in particular } \|P\Omega\| = 2\|\Omega\| \text{ for } (M, d) = (12, 3). \quad (2.3)$$

3. Covering theorem for periodic Laplacians

Theorem 1 (pull-back of cycle Laplacians). Let $M, d \geq 1$ and P, V_d as in §2.

(i) **Intertwining.** $\Delta_M P = P \Delta_d \iff d \mid M$ (for all positive M, d).

(ii) **Invariance classification (range $1 \leq d \leq M$).** V_d is Δ_M -invariant $\iff d \mid M$ or $(M, d) = (3, 2)$. (For $d > M$, P has rank M , so $V_d = \mathbb{C}^M$ is trivially invariant, though P is not an isometric embedding and the intertwining still fails unless $d \mid M$.)

(iii) **Restriction is Δ_d , not merely similar to it.** For $d \mid M$, $Q^*Q = I_d$ and $Q^*\Delta_M Q = \Delta_d$; the compressed integer form is $P^*\Delta_M P = q\Delta_d$. For $(M, d) = (12, 3)$: $Q^*\Delta_{12} Q = L_3$ and $P^*\Delta_{12} P = 4L_3$. (Q itself is not integer; the *compressed* matrix is.)

(iv) **Fourier form.** For $d \mid M$, $V_d = \text{span}\{e_j : j \equiv 0 \pmod{q}\}$; pull-back sends the $\mathbb{Z}_d$ mode j' to the $\mathbb{Z}_M$ mode qj' ; eigenvalues agree since $-4\sin^2(\pi qj'/M) = -4\sin^2(\pi j'/d)$.

The linear-algebra statements (i), (iii), (iv) hold for **all positive sizes**, including $d \in \{1, 2\}$; the graph-covering language of §4 is stated for $d \geq 3$ (or via darts, §4.3).

Background. The **divisor-case quotient construction** in Theorem 1 — the forward intertwining $\Delta_M P = P\Delta_d$ for $d \mid M$ and the compression $Q^*\Delta_M Q = \Delta_d$ — is a cycle specialization of the standard equitable-partition and network-quotient framework [GR01, OYSB13, SGP03, DL15]. The **exact converse** ($\Delta_M P = P\Delta_d \Rightarrow d \mid M$) and the **invariance classification** (invariant $\Leftrightarrow d \mid M$ or $(M, d) = (3, 2)$) are calculated here: the literature supports the forward quotient machinery, not necessarily the complete iff converse plus the exceptional $(3, 2)$ classification, and no novelty or priority is claimed. The statement is given with a self-contained proof to fix notation and to make the exact integer form $P^*\Delta_M P = q\Delta_d$ explicit. The incidence argument below is the equitable-partition (balanced-coloring) verification made concrete for cycles.

Proof of (i) — equitable-partition / incidence argument. Attach to each vertex n two directed incidences, toward $n+1$ and toward $n-1$, retained even when endpoints coincide. Suppose $d \mid M$. Partition vertices by residue mod d ; every vertex in class a has incidence destinations $a+1$ and $a-1 \pmod{d}$. Hence for the residue-indicator columns of P the aggregate adjacency row is independent of the representative n and equals the corresponding row of the d -cycle adjacency (with multiplicity); subtracting $2I$ gives $\Delta_M P = P\Delta_d$. Conversely, if $d \nmid M$ write $r = M \bmod d$, $1 \leq r \leq d-1$; comparing row 0 of $\Delta_M P$ and $P\Delta_d$ leaves the nonzero functional $u_{r-1} - u_{d-1}$ (choose u supported at $d-1$), so the identity fails. For $d > M$ pick u supported at the unseen coordinate M : $Pu = 0$ while $P\Delta_d u \neq 0$. Hence $\Delta_M P = P\Delta_d \Leftrightarrow d \mid M$. ■

Proof of (ii) — analytic converse (boundary-row cases). Divisor pairs: $V_d = \text{im } P$ is invariant by (i) ($\Delta_M P = P\Delta_d$ gives $\Delta_M \text{im } P \subseteq \text{im } P$). $d > M$: $\text{rank } P = M$ so $V_d = \mathbb{C}^M$. Conversely, if $1 \leq d \leq M$ and $d \nmid M$ (so $d \geq 2$), write $r = M \bmod d \in [1, d-1]$; compare the two same-fibre vertices 0 and d .

- If $M \geq d+2$, vertex d is interior, and $(\Delta_M Pu)_0 = u_1 + u_{r-1} - 2u_0$ while $(\Delta_M Pu)_d = u_1 + u_{d-1} - 2u_0$. Since $r-1 \neq d-1$ these differ; choosing u supported at $d-1$ gives unequal values on one fibre, so $\Delta_M Pu \notin \text{im } P$: **not invariant**.
- If $M = d+1$, both 0, d are at the wrap; the rows become $-u_0 + u_1$ and $-u_0 + u_{d-1}$, equal for all u iff $d = 2$, i.e. $(M, d) = (3, 2)$, where directly $\Delta_3(a, b, a) = (b-a, 2a-2b, b-a) \in \text{im } P$: **invariant**. For $d > 2$ pick $u_1 \neq u_{d-1}$: not invariant.

These exhaust $1 \leq d \leq M$, giving invariance $\Leftrightarrow d \mid M$ or $(M, d) = (3, 2)$. At $(3, 2)$ the induced operator in (a, b) coordinates is $\begin{pmatrix} -1 & 1 \\ 2 & -2 \end{pmatrix} \neq \Delta_2$, consistent with the intertwining failing there. ■

Proof of (iii)–(iv). From (i), $\Delta_M Q = Q\Delta_d$; left-multiplying by Q^* and using (2.2) gives $Q^*\Delta_M Q = \Delta_d$, an equality of $d \times d$ matrices with the integer entries of Δ_d (equivalently $P^*\Delta_M P = q\Delta_d$). For (iv), $Pe_j^{(d)}(n) = e^{2\pi i j'(n \bmod d)/d} = e^{2\pi i (qj')n/M} = e_{qj'}^{(M)}(n)$, so P carries $\mathbb{Z}_d$ -modes onto $\mathbb{Z}_M$ -modes indexed in $q\mathbb{Z}_M$, with equal eigenvalues by (2.1). ■

These statements are analytic. A finite regression check (paperA_proof_audit_v0_2.py) corroborates them: the intertwining holds with exact zero error for all divisor pairs $M \leq 90$; it fails for every non-divisor ≤ 60 ; the invariance classification via the correct test $(I - PP^+)\Delta_M P = 0$ matches “ $d \mid M$ or $(3, 2)$ ” over $1 \leq d \leq M \leq 40$ with no mismatch; $Q^*Q = I_3$, $Q^*\Delta_{12} Q = L_3$,

$P^*\Delta_{12}P = 4L_3$, and $\|P\Omega\| = 2\|\Omega\|$ hold exactly. Regression evidence corroborates; it does not replace the proofs above.

4. The $C_{12} \rightarrow C_3$ reduction

4.1 The covering

Specialize $(M, d) = (12, 3)$, $q = 4$. The residue map $p : C_{12} \rightarrow C_3$, $p(n) = n \bmod 3$, is a **graph covering**: it is a homomorphism, and it is bijective on the open neighbourhood $N(n) = \{n-1, n+1\}$ of every vertex (the two neighbours map to the two residues $\neq p(n)$). Each fibre $p^{-1}(r) = \{r, r+3, r+6, r+9\}$ has four vertices: **four sheets**.

4.2 Deck group and quotient

Every automorphism of C_{12} is $n \mapsto n+a$ or $n \mapsto a-n$. The rotations with $p \circ \tau = p$ are exactly $a \in \{0, 3, 6, 9\}$; a reflection would need $a - n \equiv n \pmod{3}$ for all n , impossible ($n = 0$ forces $a \equiv 0$, then $n = 1$ needs $-1 \equiv 1 \pmod{3}$). Hence

$$\text{Deck}(p) = \langle n \mapsto n+3 \rangle = \{0, 3, 6, 9\} \cong \mathbb{Z}_4,$$

acting transitively on each fibre (regular covering), so $C_3 \cong C_{12}/\langle n \mapsto n+3 \rangle$, quotient edges $\{01, 12, 20\} = K_3 = C_3$.

Statement	Truth	Reason
C_3 subgraph of C_{12}	FALSE	C_{12} has girth 12; adjacency cube has zero trace (no closed 3-walks)
C_3 quotient of C_{12}	TRUE	$C_{12}/\langle +3 \rangle = K_3 = C_3$
C_{12} covers C_3	TRUE	4-sheeted covering; deck group $\mathbb{Z}_4$
C_3 -dynamics on an invariant subspace of $\mathbb{C}^{12}$	TRUE	$V_3 = p^*\mathbb{C}^3 = \text{im } P$, invariant by Theorem 1

Compound statement: $C_{12} \xrightarrow{4:1} C_3$ **is a covering**, and $\mathbb{C}^{12} \supseteq V_3 = p^*\mathbb{C}^3$ **carries** $\Delta_{12} \upharpoonright_{V_3} = L_3$.

Terminology crosswalk. For a regular cover of an undirected cycle the following describe the *same* object and coincide for the specific regular cycle covers considered here: a **graph covering** (4-sheeted, deck group $\mathbb{Z}_4$), a surjective **graph fibration** [BV02, DL15, NRS16], a **balanced-coloring** / **synchrony-subspace quotient** [SGP03, GST05, GS06], and an **equitable-partition** / **quotient-(divisor-)matrix** reduction [GR01, OYSB13]. A synthesis tying groupoids, fibrations, and balanced colorings together is [GFBC24]. We take “covering” as the primary term and cite any of these frameworks interchangeably below; the equivalence is specific to this regular-cycle setting and is not claimed in general.

4.3 Degenerate sizes (darts)

The graph-covering language above uses ordinary open neighbourhoods and is valid for $d \geq 3$; in particular $C_{12} \rightarrow C_3$ is an ordinary simple-graph covering with no qualification. For C_1 (loop) and C_2 (doubled edge) one uses a dart/incidence description: darts $(n, +)$, $(n, -)$ with reversal $(n, +) \leftrightarrow (n+1, -)$, and dart map $(n, \pm) \mapsto (n \bmod d, \pm)$; for $d \mid M$ this respects reversal and is bijective on the two darts at each vertex, giving a multigraph covering. The linear-algebra theorem (§3) does not depend on adopting graph terminology at these degenerate sizes.

5. Fourier and symmetry decomposition

Background. The spectral splitting below is standard covering-graph / lift spectral theory. In general a regular cover's spectrum decomposes over the **irreducible representations** of the deck group; for the **abelian** regular cover considered here ($\mathbb{Z}_4$) those representations are one-dimensional characters $e^{\pi i j / 2}$, so the deck-fixed modes span the pull-back sector and inherit the base eigenvalues [GR01, DF19]. The explicit C_{12} values are worked out here for completeness.

By Theorem 1(iv) with $q = 4$,

$$V_3 = \text{span}\{e_0, e_4, e_8\}, \quad \text{spec}(\Delta_{12} \upharpoonright_{V_3}) = \{0, -3, -3\} = \text{spec } L_3.$$

These are exactly the deck-fixed modes: $e_j(n+3) = e^{\pi i j / 2} e_j(n)$, and $e^{\pi i j / 2} = 1 \iff j \in \{0, 4, 8\}$. The orthogonal complement carries the **exact** spectrum

$$\{-2 + \sqrt{3} (\times 2), -1 (\times 2), -2 (\times 2), -2 - \sqrt{3} (\times 2), -4 (\times 1)\},$$

(the values $-4 \sin^2(\pi j / 12)$ for $j \in \{1, 2, 3, 5, 6, 7, 9, 10, 11\}$; the smallest-magnitude value is exactly $-2 + \sqrt{3}$, not the rounded -0.268).

Symmetry. Δ_{12} commutes with the dihedral group D_{12} of order 24 (rotation $n \mapsto n+1$, reflection $n \mapsto -n$). Restricted to V_3 these act as the symmetries of L_3 : the rotation induces the cyclic 3-permutation of C_3 , the reflection $n \mapsto -n$ induces the transposition $(1\ 2)$ on residues $0, 1, 2$ (it sends $(0, 1, 2) \mapsto (0, 2, 1)$; the *different* reflection $n \mapsto 2 - n$, which fixes vectors (a, b, a) , induces $(0\ 2)$ — see §6.4), and complex conjugation preserves V_3 while swapping $e_4 \leftrightarrow e_8$ (an antilinear operation on the complex function space, not a deck transformation). The image of D_{12} on V_3 is the full $S_3 \cong D_3$ that stabilizes L_3 ; the kernel of the quotient action is the four-element deck group.

The one-step C_{12} rotation induces the one-step base C_3 rotation under the quotient. Separately, the order-three subgroup generated by R^4 (rotation by 4) maps isomorphically onto the base cyclic rotation ($p(n+4) = p(n)+1$), giving a subgroup lift of the same cyclic action; together with a reflection it realizes a D_3 subgroup with the same image. None of this makes C_3 a subgraph of C_{12} (§4).

6. Nonlinear extension for every M divisible by three

6.1 The ring map, with an explicit zero-amplitude convention

Adopt the site-independent, deterministic phase convention

$$\text{Arg}_0(z) = \begin{cases} \arg z, & z \neq 0, \\ 0, & z = 0, \end{cases} \quad (6.0)$$

and define the M -ring map $F_M : \mathbb{C}^M \rightarrow \mathbb{C}^M$ by

$$\Omega \mapsto \tilde{\Omega} = \Omega + \varepsilon \Omega \odot (k - |\Omega|^2) + g \Delta_M \Omega, \quad (6.1a)$$

$$(F_M \Omega)_n = \tilde{r}_n \exp\left(i \left[\text{Arg}_0(\tilde{\Omega}_n) + \lambda \sum_{m \sim n} \sin 3(\text{Arg}_0(\tilde{\Omega}_m) - \text{Arg}_0(\tilde{\Omega}_n)) \right]\right), \quad (6.1b)$$

where $\tilde{r}_n = |\tilde{\Omega}_n|$, $m \sim n$ ranges over the two ring neighbours $\{n-1, n+1\}$, and $k \in \mathbb{R}^M$ is **three-periodic** ($k_{n+3} = k_n$). When $\lambda = 0$, S_M is defined directly as the identity (no phase is invoked). The synchronization changes phases only and preserves every amplitude.

Consequences of (6.0), stated explicitly:

- F_M is **globally defined** on $\mathbb{C}^M$.
- For $\lambda \neq 0$ the extension is **generally discontinuous** at states with zero pre-synchronization components (a vanishing $\tilde{\Omega}_n$ has no genuine phase, yet its assigned value 0 enters neighbours' updates; e.g. $\tilde{v}_\delta = (\delta e^{i\alpha}, 1, 1)$ has, as $\delta \rightarrow 0$, a nonzero neighbour tending to $e^{i\lambda \sin 3\alpha}$, whose limit depends on α). Arg_0 is a *mathematical* convention distinct from IEEE signed-zero behaviour in floating point.
- The **exact pull-back identity (Prop. 2) still holds**, because the same deterministic convention is applied at corresponding residue copies.
- **Jacobian/Floquet statements are restricted to orbits whose pre-synchronization components are all nonzero**. Every representative orbit used in §7–§8 satisfies this comfortably (minimum pre-sync amplitude > 0.948).

For $M = 3$ the two neighbours $\{n-1, n+1\}$ of any node are the two other nodes, so S_3 is the all-pairs synchronization of §1 and F_3 is the reference three-node map.

Background (coupling family). The phase term $\sin 3(\phi_i - \phi_j)$ is a higher-harmonic phase coupling. Higher-harmonic coupling occurs in phase-oscillator clustering studies [HMM93], with symmetry-based phase-locking context in [AS92]. Here the three residue classes are the network partition whose invariance is verified in Proposition 2; we do not derive the cluster count from those citations, which concern specific (first / second) harmonics rather than a general all- m theorem. This places the *coupling family* only; it is not itself novel, and it is not a claim about the specific combined map, whose closest dynamical neighbours are discussed in §6.5.

6.2 Exact reduction for all $M = 3q$

Proposition 2. Let $M = 3q$, $q \geq 1$, and k three-periodic. With the convention (6.0),

$$F_M(P\Omega) = PF_3(\Omega) \quad \text{for all } \Omega \in \mathbb{C}^3,$$

so $V_3 \subset \mathbb{C}^M$ is F_M -invariant and $F_M \upharpoonright_{V_3} \cong F_3$.

Proof. Let $f = P\Omega \in V_3$. *Pointwise term:* $k = Pk^{(3)}$ and entrywise operations act residue-classwise on pulled-back vectors, so $\varepsilon f \odot (k - |f|^2) = P(\varepsilon \Omega \odot (k^{(3)} - |\Omega|^2))$. *Coupling:* $g\Delta_M(P\Omega) = P(g\Delta_3\Omega) = P(gL_3\Omega)$ by Theorem 1(i). Hence $\tilde{f} = P\tilde{\Omega} \in V_3$ with $\tilde{\Omega} = \Omega + \varepsilon \Omega \odot (k^{(3)} - |\Omega|^2) + gL_3\Omega$. *Synchronization:* $\tilde{f} \in V_3$, and for every n (including the wrap), divisibility gives the modular identity

$$\{((n-1) \bmod M) \bmod 3, ((n+1) \bmod M) \bmod 3\} = \mathbb{Z}_3 \setminus \{n \bmod 3\}. \quad (6.2)$$

So the two ring-neighbour contributions equal the two all-other-node contributions of S_3 evaluated at residue $n \bmod 3$. All increments use the single pre-synchronization vector $\tilde{f}$ (a **simultaneous**, not sequential, update), and amplitudes are restored unchanged; hence $F_M(P\Omega) = PF_3(\Omega)$. At any zero pre-sync component the common convention (6.0) is applied identically at all copies of the residue, preserving the identity. ■

Background. Balanced synchrony and quotient dynamics are established for admissible vector fields [SGP03, GST05, GS06], and explicitly for smooth network maps in [NRS16]; graph-fibration semiconjugacy is the continuous-time background [DL15]. Proposition 2 verifies the corresponding compatibility for this composed update, including the shared zero-amplitude convention (6.0) — whose discontinuous-at-zero Arg_0 handling is proved here, not inherited from

those sources. The model-specific ingredient is the elementary verification that the harmonic-3 neighbour term is admissible for this ring — the neighbour-residue identity (6.2), the local-bijectivity / balance condition made explicit. As in §1.3, P is a bijection onto V_3 , so the identity says F_3 equals $F_M \upharpoonright_{V_3}$; we avoid the looser word “conjugacy.”

The modular identity (6.2) holds for **every** M divisible by three, $M = 3$ included — there is nothing special about $M = 12$ (contrary to a claim in an earlier draft). The argument uses simultaneous updates; a sequential/in-place phase sweep, using already-modified neighbours, defines a different map to which it does not apply. The reference synchronization is simultaneous (all pairwise differences are formed from one phase vector), so it matches.

Two distinct scopes, to be kept separate:

- **Exact invariance / reduction:** every M divisible by three.
- **Transverse-stability atlas (§7–§8):** $M = 12$ only. Larger rings have different transverse modes and are not studied here.

6.3 Numerical cross-check (one-step identity, and a long-trajectory caveat)

The reference F_3 is a specific implementation, so we confirm the reduction against it. Independent checks (referee reproduction, direct cyclic matrices, simultaneous phase updates, source loaded only after re-derivation):

test	error measure	value
one-step $F_{12}(P\Omega)$ vs lifted reference, 50 runs (seeds 3,17,101,2026,8675309)	max per-component	5.53×10^{-15}
free 600-step trajectories, defaults, 5 seeds	max per-component	2.13×10^{-15}
free 600-step trajectories, $(g, \lambda) = (0.2, 0.5)$, 5 seeds	max per-component	3.82×10^{-15}
one-step generalization, $M = 3, 6, 9, \dots, 30, 63$	max per-component	2.61×10^{-16}
free 600-step over all random parameter cases	max per-component	4.94; Euclidean max 11.09

The error measure is the maximum per-component (infinity-norm) discrepancy, **not** a Euclidean norm. The last row is stated deliberately: **an exact one-step algebraic identity does not guarantee indefinitely close floating-point trajectories in unrestricted dynamical regimes** — tiny implementation-order differences are amplified in sensitive parameter regions (the worst case here is $g \approx 0.0076, \lambda \approx 0.637$, whose one-step discrepancy is nonetheless bounded by the first row). This forward divergence is a numerical-analysis caveat, not a refutation of the exact algebra; the selected default and high- λ regimes do reproduce closely. The read-only import of the reference map wrote no bytecode to the source tree.

6.4 The exceptional $(M, d) = (3, 2)$ case

Theorem 1(ii) admits one non-divisor invariant image: $(3, 2)$. Here $2 \nmid 3$, so $p(n) = n \bmod 2$ is **not** a covering, and $V_2 = \text{im } P = \{(a, b, a)\}$ is not a pulled-back C_2 -sector; it is nonetheless the fixed space of the node reflection $n \mapsto 2 - n$ (the transposition $(0 \ 2)$, which fixes vectors (a, b, a) ; distinct from the $n \mapsto -n$ reflection of §5, which is $(1 \ 2)$ of L_3 , and L_3 -invariant for that symmetry reason. This illustrates that **invariant subspaces of Δ_M can arise from symmetry without a covering**; only the $d \mid M$ subspaces are the pull-back sectors of Theorem 1. The induced operator there is $\begin{pmatrix} -1 & 1 \\ 2 & -2 \end{pmatrix} \neq \Delta_2$, consistent with intertwining failing.

6.5 Closest dynamical neighbours (context)

The genre of the full map is a **complex coupled map with cluster states**. The nearest cited neighbours are Kaneko's globally coupled maps [Kaneko90] (discrete-time cluster states, but with real logistic maps rather than a complex Mexican-hat amplitude step plus a *separate* harmonic-3 phase synchronization), and the discrete-time network-map framework [NRS16]; Stuart–Landau / Ginzburg–Landau ring models are a continuous-time relative of the amplitude part. **We did not locate a near-identical model in the focused literature search** — the particular combination of a complex Mexican-hat amplitude nonlinearity, cycle-Laplacian coupling, and a separate harmonic-3 phase step was not found as a studied model. We record this as an *unresolved* novelty status, not a novelty claim (§12; PAPER_A_LITERATURE_CONTEXT_v0.1.md).

7. Transverse stability ($M = 12$)

We ask whether V_3 is not merely invariant but attracting for $M = 12$: do nearby twelve-dimensional states collapse onto the reduced sector?

7.1 Real deck-character block structure

Over $\mathbb{R}$ (identify $\mathbb{C}^{12} \cong \mathbb{R}^{24}$), let T be the realified deck generator (translation by three). At a nonsingular on-sector point x ($Tx = x$), differentiating the deck **equivariance** $F_{12}(Tx) = TF_{12}(x)$ gives $DF_{12}(x)T = TDF_{12}(x)$; equivariance (not mere invariance of V_3) is what forbids normal-to-tangent mixing. Define the real orthogonal projectors

$$E_0 = \frac{1}{4}(I + T + T^2 + T^3), \quad E_2 = \frac{1}{4}(I - T + T^2 - T^3), \quad E_{13} = I - E_0 - E_2,$$

with ranks 6, 6, 12, giving

$$\mathbb{R}^{24} = U_0 \oplus U_2 \oplus U_{13}, \quad U_0 = \text{range } E_0 = \text{realification of } V_3.$$

U_0 is **tangential** (reproduces the three-node dynamics); $U_2 \oplus U_{13}$ is the 18-dimensional **transverse** (normal) space whose real Jacobian block decides transverse stability. Over $\mathbb{C}$ the i and $-i$ characters are separate eigenspaces linked by reality; **in the real Jacobian they combine into the single block U_{13}** (a real-linear derivative can mix the complex Fourier sectors 1 and 3, so they are not individually invariant — e.g. at $\Omega = (1, \dots, 1)$, $g = \lambda = 0$, the cubic derivative sends a character-1 unit vector to a character-3 component of norm 0.05). The projectors are orthogonal, so there is no first-order normal/tangent leakage (measured $\lesssim 10^{-14}$).

Background. The original master-stability framework — block-diagonalizing the variational dynamics transverse to the synchronous state — provides the background [PC98]; the symmetry-adapted / irreducible-representation cluster decomposition that this deck-character block-diagonalization instantiates (the abelian $\mathbb{Z}_4$ case) follows the framework of [Pecora14, Sorrentino16]. The method is standard; only the explicit real-character pairing U_{13} and the numeric blocks below are specific to this ring.

7.2 Linear spectral test vs nonlinear attraction

Two regimes are distinguished; conflating them is an error we avoid.

- **Fixed point.** For a fixed point of F_3 , the transverse Jacobian $J \upharpoonright_{U_2 \oplus U_{13}}$ has moduli deciding the **linear** verdict.
- **Period- p orbit.** The correct object is the **normal Floquet multiplier** $\rho_\perp$, the spectral radius of the transverse monodromy $\prod_{s=1}^p J_s \upharpoonright_{U_2 \oplus U_{13}}$; per-step factor $\rho_\perp^{1/p}$. Fixed-point (single-step) language must not be used for periodic regimes.

The linear test is not an iff criterion for nonlinear attraction. Precisely:

- $\rho_{\perp} < 1$: the normal variational dynamics are exponentially asymptotically stable (**linearly transversely stable**); the orbit's transverse linearization decays exponentially.
- $\rho_{\perp} > 1$: the orbit is **linearly transversely unstable**.
- $\rho_{\perp} = 1$: the linear test alone is **inconclusive** about nonlinear attraction.

We therefore do **not** write "attracting iff all multipliers < 1 ", and we avoid the word *contractive* (which would require a proved operator norm < 1 in a stated norm). Two facts make the distinction concrete. (a) Within this map family, at $k = 0, g = \lambda = 0, \varepsilon > 0$, each site obeys $z^+ = z(1 - \varepsilon|z|^2)$: the derivative at 0 is the identity (every multiplier equals 1), yet all small amplitudes decrease monotonically to 0 at the **algebraic** rate $|z_n| \sim (2\varepsilon n)^{-1/2}$ — attraction without exponential rate, unresolved by the linear test. (b) Spectral radius < 1 does not imply Euclidean per-step contraction: the stable $g = 0.2, \lambda = 0.5$ period-2 orbit has normal monodromy spectral radius 0.9912953186 but Euclidean operator norm 1.0326263720 > 1 in an orthonormal normal basis, so some perturbations grow transiently over one period before decaying. Our terminology is accordingly: *linearly transversely stable, normal variational decay, numerically observed transverse attraction*.

7.3 The default attractor

At defaults the reduced trajectory is a fixed point in V_3 . The tangential (U_0) moduli and the transverse (U_2, U_{13}) moduli are (independently recomputed):

$$U_0 : (0.07264, 0.08864, 0.33013, 0.42951, 0.68095, 1), \quad U_2 : (0.11947, 0.17736, 0.47305, 0.48865, 0.73101, 0.85127),$$

with largest U_{13} modulus 0.94892. Every **transverse** modulus ($U_2 \cup U_{13}$) is < 1 : V_3 is **linearly transversely stable** at defaults. The single unit modulus lies in the tangential block U_0 and is the neutral global-phase direction of the three-node system itself (not a transverse direction). (The proof-audit companion labels this list "tangential moduli", not "transverse".)

7.4 Representative regimes ($M = 12$)

V_3 is linearly transversely stable over a substantial sampled region, not everywhere. Six regimes ($\rho_\perp = p$ -step normal Floquet radius; $\chi = \|\operatorname{Re} \Omega \times \operatorname{Im} \Omega\|$; independently reproduced at seeds 0,19,407):

g	λ	type	p	χ	$\rho_\perp$	$\rho_\perp^{1/p}$	linear verdict
0.2	0.001	fixed	1	≈ 0	0.948919213	0.948919213	stable (default)
0.2	0.5	period-2	2	1.558108	0.991295319	0.995638146	stable (high- χ ; op-norm > 1 , transient growth)
0.1	0.3	period-2	2	1.677188	1.134220618	1.064997943	unstable
0	0.4	period-8	8	2.66357	2.416706689	1.116613805	unstable
0.3	0.5	fixed	1	≈ 0	1.094226451	1.094226451	unstable
0	0	fixed, free phases	1	initial-dependent	1	1	neutral (linear test inconclusive)

Neither high χ nor stronger synchronization alone predicts transverse stability: instability occurs at $\chi \approx 0$ (the $g=0.3, \lambda=0.5$ fixed point) and stability coexists with sizable χ (the $g=0.2, \lambda=0.5$ orbit). The neutral case requires $g = \lambda = 0$: setting $g = 0$ alone does *not* decouple phases or imply neutrality — the λ -dependent phase coupling remains, and the table's own $g=0, \lambda=0.4$ row is a transversely unstable period-8 orbit ($\rho_\perp = 2.4167$). At $g = 0$ different phase basins can realize different χ without changing the period or normal multiplier; that is not decoupling.

7.5 A fixed-branch crossing, and regional (not global) scope

On the positive in-phase fixed branch there is a transverse instability onset. Solving $\varepsilon r_i(k_i - r_i^2) + g(\Delta_3 r)_i = 0$ for the positive in-phase radii (crossing solution $\approx (1.62613, 1.63733, 1.93169)$, residual 2.3×10^{-16}) and setting the lowest eigenvalue of the projected normal radial operator to -1 gives

$$g_* = 0.4220744431784353 \quad (\text{canonical } k, \varepsilon = 0.05, \lambda = 0.001),$$

a **transverse -1 eigenvalue crossing in the real character-2 block (U_2) of the selected positive in-phase reduced fixed branch**, near the $\lambda = 0.001$ slice (eigenvalue slope ≈ -3.97 , nonzero; a full-map finite-difference normal radius at the crossing is 0.99999999999996 ; nearby sampled reduced orbits remain on that branch to $< 8.1 \times 10^{-16}$). It is not a longitudinal loss mislabeled as transverse, and it is not a basin claim. The separately located stationary-branch crossing $\lambda \approx 0.42844590$ at $g = 0.2$ occurs on a branch already longitudinally unstable there (largest tangential modulus 1.238 , sampled attractor period-2), and is **not** promoted to an attractor boundary.

Terminology. We use the description “transverse -1 crossing / loss of transverse stability” (a transverse period-doubling-type instability of the synchrony sector). We deliberately do **not** re-label it a *blowout* or a *bubbling* / *riddling* bifurcation: the blowout setting [OS94] and the distinct bubbling / riddling setting [ABS94] both concern a **chaotic** invariant set losing transverse stability, which is not the fixed-branch situation studied here. The plain description already fits and is retained.

Atlas evidence (authoritative rows; one coarse grid with adaptive refinement, **not** two independent grids or area fractions):

stability label	coarse grid (190)	all samples incl. coarse (379)
stable	96	150
unstable	40	132
boundary / neutral	1	1
unresolved	53	96

Adaptive refinement concentrates samples at transitions, so $150/379$ is **not** the area fraction of a stable region; we describe a *substantial sampled stable region* only, with 32 stable samples at $\chi > 0.1$. The 37 retained edge brackets are **numerical evidence for transitions, not proof of smooth codimension-one boundaries everywhere**. At the numerically located g_* crossing, the computed eigenvalue slope is approximately -3.97 and is nonzero to the reported numerical precision (the root tolerance is numerical precision, not an interval-certified error bound). No global-basin or complete-bifurcation claim is made; unresolved samples remain unresolved.

8. Numerical perturbation verification ($M = 12$)

Background. Near-cluster numerical evolution is illustrated in [Schaub16, §III.B.2 and Fig. 3]. The normal perturbations and measured decay / growth rates below are our own tests of this map; no citation is needed to justify the measured rates.

Lift a reduced state and add a transverse perturbation, $\Omega^{12} = P\Omega^3 + \varepsilon_\perp \tilde{\zeta}$, $\tilde{\zeta} \in U_2 \oplus U_{13}$ (real 18-dimensional normal space), advance the full F_{12} (never forced back into V_3), and measure $d_\perp(t) = \|(I - QQ^*)\Omega(t)\|$. Independently initialized at a retained orbit point plus a dominant normal Floquet direction, amplitudes $10^{-7}, 10^{-5}$, 1800 steps, period-stroboscopic fits restricted to $10^{-10} < d_\perp < 10^{-4}$ and to distances $\geq 100\times$ the simultaneously advanced control's sector

error (saturation and roundoff-floor samples excluded). Measured normal log-rates match the variational prediction:

g, λ	perturbation	predicted log-rate	measured log-rate	fitted step range
0.2, 0.001	10^{-5}	-0.05243161232	-0.05243161307	8–207
0.2, 0.5	10^{-5}	-0.004371394220	-0.004371394222	8–406 (every 2nd)
0.1, 0.3	10^{-7}	+0.06297286724	+0.06297286730	8–108 (every 2nd)
0, 0.4	10^{-7}	+0.11030071763	+0.11030071784	16–56 (every 8th)
0.3, 0.5	10^{-7}	+0.09004767578	+0.09004767572	8–76
0, 0	10^{-7}	≈ 0	≈ 0	8–207

Eleven of the twelve runs give an admissible fit; the largest rate error across those fits is 1.64×10^{-8} . The period-8 run at amplitude 10^{-5} has only one usable unsaturated stroboscopic sample under the strict cutoff, so no slope is reported from it (its 10^{-7} companion supplies the check). The two stable regimes ($g=0.2, \lambda=0.001$ and $0.2, 0.5$) show **negative** rates (normal decay); the unstable regimes show positive rates and depart into off-sector attractors. Unperturbed lifted controls remain in V_3 to $\sim 10^{-15}$: the exact invariant sector does not spontaneously break in exact arithmetic, and infinitesimal transverse errors grow precisely where the transverse linearization predicts.

These are the corrected baseline perturbation values; an earlier draft mixed the baseline and a later feedback experiment (different amplitudes, durations, and floors), and reversed two rate columns. The two experiments are kept separate here, and every figure is drawn from a named data row with its parameters, seed, amplitude, direction, duration, and error definition (Appendix D).

9. Relation to a second ring operator

A second, independently defined operator carries the same L_3 on its three-periodic phase sector: a spectral-branching ("RSB") phase Laplacian built directly from Δ_{12} . Its full state space is the tensor product

$$\mathbb{C}_{\text{channel}}^3 \otimes \mathbb{C}_{\text{phase}}^{12} \otimes \mathbb{C}_{\text{helicity}}^2$$

on which channel and helicity operations act on spectator factors and the phase Laplacian is Δ_{12} . The **preserved sector** of the default update ($\alpha = 0$) is accordingly $\mathbb{C}^3 \otimes V_3 \otimes \mathbb{C}^2$; the phase Laplacian commutes with the residue pull-back, so on nonzero admissible states the full Hamiltonian intertwines with the $M = 3$ Hamiltonian under the lifted phase map. Because the source normalizes its state, a full **normalized**-step identity uses the isometry Q on the phase factor (i.e. $Q \otimes \text{id}$ factors), not an unscaled nonlinear P -identity; and the domain excludes the zero state, where a normalization fallback (raising without an RNG, or a random RNG fallback) produces a generically non-three-periodic normalized state. The optional dominant-index contraction ($\alpha > 0$) selects a single band via a one-hot mask, breaking the deck symmetry — hence breaking the sector — including under adaptive entropy scaling.

Discipline: this is **shared exact algebra** (both ring operators contain L_3 as their three-periodic sector because both are built from Δ_{12} , exactly as Theorem 1 predicts), **not** an identity between the three-complex-variable oscillator map and the 72-complex-component RSB evolution. They never share a state vector; nothing more is asserted.

Background. The shared phase-Laplacian sector uses the standard quotient / synchrony mechanism [GR01, SGP03]; §9 establishes the particular comparison for the two operators considered

here. The literature supports the mechanism, not an unrestricted full-operator theorem or any RSB-specific result, and nothing beyond the shared quotient algebra of this pair is claimed.

10. Feedback compatibility (sufficient condition)

Suppose feedback modifies **only** the amplitude coefficient k , through an effective coefficient depending on read-out/auxiliary data.

Corollary 3 (sufficient compatibility condition for k -only feedback). Suppose the auxiliary/read-out variables admit compatible lifts such that on V_3

$$k_{\text{eff}}^M(P\Omega, \eta_M) = P k_{\text{eff}}^3(\Omega, \eta_3), \quad (10.1)$$

where inputs are typed as follows: either an oscillator argument $x \in \mathbb{C}^3$ with a compatible lifted state/read-out, or a **scalar** read-out z with equal scalar read-outs on lifted states (in which case P is **not** applied to z). Then V_3 remains invariant under the feedback-augmented F_M , and $F_M(P\Omega) = PF_3(\Omega)$ continues to hold.

Proof. The only change to F_M is $k \rightsquigarrow k_{\text{eff}}^M(P\Omega, \eta_M)$ in (6.1a). By (10.1) this equals $Pk_{\text{eff}}^3(\Omega, \eta_3)$, itself three-periodic and pulled back; the proof of Proposition 2 applies verbatim. ■

Background. Corollary 3 is a specialization of the standard principle that a coupling or modification respecting the balanced (equitable) partition preserves the synchrony subspace [SGP03, Schaub16]. The k -only condition (10.1) is the explicit, model-specific form of that principle for this architecture; no stronger general theorem is needed, and it is stated as sufficient, not necessary (caution 1).

Three cautions keep this in proportion.

1. **Sufficient, not iff.** (10.1) is a sufficient one-step condition for this architecture; it is **not** necessary. Counterexample to a necessity claim: with $k_{\text{eff}}^M = Pk_{\text{eff}}^3 + b$ for non-periodic b and a compensating feedback term $-\varepsilon\Omega \odot b$ before synchronization, the map is unchanged so the reduction is exact although (10.1) fails; likewise arbitrary coefficients differing only on zero coordinates leave the pre-sync map unchanged. The earlier companion's "iff" wording is withdrawn.
2. **Preservation of $V_3 \neq$ equality to a specified F_3 .** A common scalar depending on the **unnormalized** norm $\|P\Omega\| = \sqrt{M/3} \|\Omega\| = 2\|\Omega\|$ (for $M = 12$; the required factor is $\sqrt{12/3} = 2$, not an "undefined sector count") preserves V_3 while generally inducing a *different* three-node feedback; invariance and equality-to- F_3 are distinct. Read-out normalization must be matched for equality.
3. **Compatibility bears on invariance only; transverse boundaries can move.** A compatible feedback can preserve the sector yet change transverse derivatives. One recovered scalar-amplitude law (all effective coefficients of each machine multiplied by one common real scalar) is compatible under a consistent lift and empirically shifts the low- λ fixed-branch boundary slightly toward instability (bracketed $g \in [0.421, 0.4215]$ vs no-feedback 0.42207) — a **local** shrinkage; across matched samples some transverse factors improve and others worsen, 167 matched comparisons retain an unresolved side, and the net global change of the stable region is **undetermined**. This section is a compatibility criterion, not a physical-portal result.

11. Discussion

The paper isolates one exact structure: a three-state nonlinear oscillator is the restriction to an invariant subspace V_3 of an M -ring map (any M divisible by three), where V_3 is the pull-back of functions along the covering $C_M \rightarrow C_3$. The linear backbone (Theorem 1) is elementary and general — $\Delta_M P = P \Delta_d \iff d \mid M$, with $Q^* \Delta_M Q = \Delta_d$. The nonlinear content (Proposition 2) rests on the modular identity (6.2), which holds for every $M = 3q$. The dynamical content (§7–§8, $M = 12$) is that V_3 is linearly transversely stable over a substantial sampled region, established by the real deck-character decomposition and confirmed by direct perturbation, with explicit unstable and neutral counterexamples and the linear/nonlinear distinction respected throughout.

Three features make the result more than a curiosity: the reduction is *exact* (one-step identity to rounding, identical in exact arithmetic); the full real linearization is *structured* by the deck group into $U_0 \oplus U_2 \oplus U_{13}$, with tangential block U_0 and transverse restriction $U_2 \oplus U_{13}$; and the same L_3 -in- Δ_{12} phenomenon recurs in a second, independently built operator (§9), via the standard quotient / synchrony mechanism [GR01, SGP03] applied to the particular pair compared there.

12. Limitations

- **The ring extension is a construction of this paper**, a 2026 mathematical object. We do **not** claim any pre-existing three-node system was historically designed as an M -ring, nor that any M is privileged beyond the arithmetic $3 \mid M$.
- **Transverse attraction is local and regional** ($M = 12$), with explicit unstable and neutral counterexamples; no global basin, complete bifurcation diagram, or longitudinal-stability theorem is claimed. Unresolved samples remain unresolved. At $\rho_\perp = 1$ the linear test is inconclusive.
- **No physical interpretation.** Ω, z, χ, k , and the graphs are abstract; no physical scale, particle, spacetime, or measured phenomenon is claimed.
- **No systems claim.** §9 asserts only shared algebra between two mathematical operators, not dynamical identity, and no production/memory-system claim.
- **Feedback compatibility is a sufficient criterion**, not a portal result; the recovered scalar law's global stability effect is undetermined.
- **No novelty/priority claim.** §§3–5 are elementary standard theory — network quotients, graph fibrations, equitable partitions, synchrony subspaces, master-stability / cluster-synchronization (§1.2–1.4; PAPER_A_LITERATURE_CONTEXT_v0.1.md). The model-specific content is the §§6–10 constructions and the §7–§8 transverse-stability atlas, whose novelty is recorded as **unresolved**, never asserted; a literature-context pass is not a novelty determination.

13. Conclusion

Strongest supportable conclusion: **for every M divisible by three, the specified three-state nonlinear map admits an exact invariant pull-back realization on the M -cycle with three-periodic coefficients; for the $M = 12$ realization studied numerically here, the invariant sector exhibits linearly stable and unstable transverse regimes, and direct nonlinear perturbations confirm transverse attraction in representative linearly stable regimes.** The reduction is a graph covering $C_M \rightarrow C_3$ with pull-back V_3 , on which the ring Laplacian equals the triangle Laplacian and the ring nonlinear map equals the three-node map; for $M = 12$ the full real linearization block-decomposes by the deck group $\mathbb{Z}_4$ into $U_0 \oplus U_2 \oplus U_{13}$, with tangential block U_0 and transverse restriction $U_2 \oplus U_{13}$ linearly stable over a substantial sampled region. We claim nothing about nature, quantum mechanics, or the historical origin of the three-node map.

Appendix A — Theorem proofs (supplementary)

A.1 Sizes. (2.1) follows from circulant diagonalization for $M \geq 3$; the degenerate Laplacians are $\Delta_1 = (0)$ and $\Delta_2 = \begin{pmatrix} -2 & 2 \\ 2 & -2 \end{pmatrix}$. Theorem 1(i),(iii),(iv) hold at all positive sizes (P01 boundary table: $d = 1$ constant column annihilated by Δ_M ; $M = d = 2$ identity lift; etc.).

A.2 Exactness of $Q^* \Delta_M Q = \Delta_d$. From (i), $\Delta_M Q = Q \Delta_d$; left-multiply by Q^* and use $Q^* Q = I_d$. The compressed matrix $q^{-1} P^* \Delta_M P$ is integer ($= \Delta_d$); Q itself need not be integer.

A.3 Regularity/fixed space. For a regular covering the pull-back subspace equals the deck-fixed space; §5 verifies this for $(12, 3)$ via $e^{\pi i j / 2} = 1 \iff j \in \{0, 4, 8\}$.

A.4 Converse. The boundary-row argument (§3, proof of (ii)) is the analytic classification; $(3, 2)$ is the sole non-divisor invariant image in $1 \leq d \leq M$, a symmetry fixed space, not a covering sector.

Appendix B — Nonlinear reduction details

B.1 Stage-by-stage commutation. Pointwise term: residue-classwise action on pulled-back vectors. Coupling: Theorem 1(i). Synchronization: the modular identity (6.2), valid for every $M = 3q$ including the wrap. Composition gives $F_M \circ P = P \circ F_3$ (Prop. 2). Simultaneous updates are essential; a sequential sweep defines a different map.

B.2 Neighbour-residue identity. $\{((n-1) \bmod M) \bmod 3, ((n+1) \bmod M) \bmod 3\} = \mathbb{Z}_3 \setminus \{n \bmod 3\}$ for every n and every M with $3 \mid M$. This is the sole nonlinear-specific ingredient; it is **not** special to $M = 12$.

B.3 Zero-amplitude convention. (6.0) makes F_M global; the reduction is preserved because the identical deterministic convention is applied at all residue copies. Jacobian/Floquet claims use only orbits with nonzero pre-sync components.

Appendix C — Numerical methods

C.1 Regression audit (paperA_proof_audit_v0_2.py, pure math, no implementation import): 20/20 checks corroborating (not replacing) the analytic proofs — intertwining exact for all divisor pairs $M \leq 90$ and failing for every non-divisor ≤ 60 ; the **invariance classification via $(I - PP^+) \Delta_M P = 0$** matching " $d \mid M$ or $(3, 2)$ " over $1 \leq d \leq M \leq 40$; the $(5, 3)$ demonstration that the old $P^+ P = I$ check was not an invariance test; $Q^* Q = I_3$, $Q^* \Delta_{12} Q = L_3$, $P^* \Delta_{12} P = 4L_3$, $\|P\Omega\| = 2\|\Omega\|$; $V_3 = \text{span}\{e_0, e_4, e_8\}$ with exact complement spectrum $\{-2 \pm \sqrt{3}, \dots\}$; the real projectors E_0, E_2, E_{13} with ranks 6, 6, 12; the modular identity (6.2) for $M \in \{3, \dots, 63\}$; and the $z^+ = z(1 - \varepsilon|z|^2)$ counterexample (unit multiplier, algebraic decay $|z_N| = 7.07 \times 10^{-3}$ matching $(2\varepsilon N)^{-1/2}$).

C.2 Reduction check (paperA_reduction_check.py, read-only reference import, sys.dont_write_bytecode=True, zero.pyc written): one-step and free-trajectory discrepancies as in §6.3 (max per-component / infinity-norm). The referee's independent reproduction (fourth-order finite-difference Jacobian, direct cyclic matrices, source loaded only after re-derivation) gives one-step 5.53×10^{-15} , defaults 600-step 2.13×10^{-15} , high- λ 3.82×10^{-15} , generalization 2.61×10^{-16} , and the unrestricted-regime caveat (4.94 per-component / 11.09 Euclidean).

C.3 Stability/perturbation numerics. Transverse Jacobians computed analytically in radial/tangential coordinates and cross-checked against a fourth-order central-difference stencil (largest multiplier difference under h -refinement 5.20×10^{-10}); trajectory classification periods 1–64 at relative residual $< 10^{-10}$; unresolved states never assigned fixed-point Jacobians; perturbation fits stroboscopic, pre-saturation, with the strict cutoffs of §8. The stability atlas and perturbation data are the Phase-3V reproduction and the referee reproduction; this pa-

per re-derives the algebra and the reduction independently and cites those data rather than regenerating the atlas.

Appendix D — Reproducibility

Authoritative derivations and data (project-relative to `TRIOCTAGON_new/`):

- `reconstruction/EXACT_3_12_24_STRUCTURE_v0.1.md` — Theorem 1, Fourier form, tower, nonlinear extension.
- `reconstruction/CLAUDE_PHASE3_HOST_FEEDBACK_AND_3_12_24_RECOVERY_v0.1.md` — parent report.
- `reconstruction/CODEX_PHASE3V_PORTAL_AND_12RING_VERIFICATION_v0.1.md` — independent reproduction: covering/fibre/deck; reduction; transverse-stability atlas; direct perturbations; **feedback compatibility in its §9, with the feedback experiments in its §15** (corrected from an earlier miscitation).
- `reconstruction/papers/CODEX_PAPER_A_REFEREE_REPORT_v0.1.md` — the referee report driving this revision, and `verification/paperA_referee/` — its data (`referee_summary.json`, `reference_crosscheck.csv`, `preservation_result.json`).
- `reconstruction/MASTER_RESEARCH_RECONSTRUCTION_MAP_v0.1.md`, `MASTER_EQUATION_LINEAGE_v0.1.md`, `MASTER_PUBLICATION_ROADMAP_v0.1.md` — provenance/scope (Paper A entry).

Verification code (this paper): `verification/paperA/paperA_proof_audit_v0_2.py` and `verification/paperA/paperA_reduction_check.py` (copies of the publication-specific scripts; the earlier copies under `verification/master_research_map/` are retained unchanged). Parent verification `verification/phase3_host_feedback/phase3_structure_checks.py` and Phase-3V data `verification/phase3v_portal_12ring/`.

Reference implementation (read-only, definitions only): `kernel_T0/model_core.py` (`TriOctaPhaseLockModel.phase_lock_step = (6.1a); L3`), `kernel_T0/phase_triad_sync.py` (the synchronization S ; simultaneous, matching (6.1b)), `kernel_T0/rsb_model.py` (the second ring operator, §9), `kernel_T0/constants_selector.py` (the amplitude coefficient k). Full sha256 content hashes are in `verification/master_research_map/hashes.tsv` (not modified); no source file was modified, and no hard-coded container path is assumed — scripts derive the project root or take it as an argument.

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Full metadata (with DOIs and the verification note) is in the companion PAPER_A_REFERENCES_v0.2.md. All references are literature context / positioning; none supports a mathematical step of the frozen result. (Field04 and Stewart07, cited in v0.4, became unused after the v0.5 attribution patches and are removed from this publication bibliography; they remain in the reference ledger.)

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